

Sixteen passes: what was proposed, what was tested, and what survived

Ava Kim — cat-pose-benchmark, technical note 06, v0.2, 20 September 2026 (v0.2: criteria in Table 1 corrected against the pass log after the reviewer's audit; passes 17–19 added)

Repository: <https://github.com/in-c0/cat-pose-benchmark> · Follows note 05 · Licence of this note: CC-BY-4.0

Abstract

This is the audit trail behind note 05. Over one night, sixteen proposals for improving the grounded tail method were made by a reviewer model, built, tested against a written-down criterion, and kept or dropped on the result. One survived: the two-sided SAM2 bridge (v3). Everything else — denser sampling, re-weighted temporal terms, a wider candidate lattice, a lower detector threshold, a looser gate, a cheaper switch, tracking until failure, a consensus audit, two forms of zero-shot semantic verification and one of mask geometry — failed the test it was given, and the table below says which test and by how much. The frame truth was revised four times on the way, and those revisions are in the same table, because three of them were found by method output and the record should say so. The pass log with every proposal and reply in full is in the portfolio hub.

1. Why a table

Note 05 reports the outcome. It does not show how many things were tried and dropped to get there, and a reader deciding whether to try one of them again needs the negative results as much as the positive one. Each row below is one pass: what the reviewer proposed, what was built, the criterion written down before the result was known, the number, and the disposition. Where the criterion was not written down first, the row says so.

2. The passes

Table 1. The passes. Counts are TP / wrong on visible / missed / wrong on hidden on the four development clips under the truth version and scorer in force at the time, unless stated; note 05 v0.3 re-reports the surviving methods under scorer v2. "Rejected" means the method was not changed. The criteria in rows 2, 3, 5, 6a and 13 were corrected in v0.2 against the pass log after the reviewer pointed out that v0.1 had paraphrased them.

pass	hypothesis	what was built	predeclared test	result	disposition
1	denser sampling helps the temporal pass	inference at 2–4× the review rate, scored at the review timestamps	+recall, ≤ 1 new FP	+3 recall points under the old proxy; 73/1/25/8 vs 77/0/22/2 once the truth was written down	rejected; the proxy was the error
2	temporal terms should scale with frame interval	time-normalised Viterbi <i>emissions</i> (the switch cost stayed at 0.20), reviewer's weights	recall up with no new FP on either clip	<i>Cat Plays</i> collapses: 63/0/32/1	rejected
3	a wider candidate lattice recovers misses	oracle: correct box in top-5 but not top-2	≥ 4 new usable tails in K5■K2, ≥ 3 of them surviving the crop check	1 frame	not built
4	—	truth v1 frozen; scorer; rescoring of every method	—	v1 77/0/22/2 (P 0.975, R 0.778)	baseline
5	the detector threshold hides true tails	dump of every "tail" box down to 0.05 on missed and on hidden-tail frames	≥ 4 usable boxes below 0.20 on missed frames, ≥ 3 of them at ≥ 0.10	2 usable; 2.9 junk boxes ≥ 0.20 per hidden frame	rejected
6a	the cat gate is too tight	30 % containment instead of the centre rule	none predeclared — a diagnostic I had already started	identical to v1	rejected
6b	a cheaper switch lets tails in	switch cost 0.20 → 0.15/0.10/0.05/0.00	a plateau before the first FP	first FP arrives with the first TP	rejected
6c	two-sided propagation can fill a gap safely	left / right / union / intersection bridges, gaps ≤ 3	≥ 2 TP, 0 FP over v2	intersection +4 TP, 0 FP; the others admit a 3× blob	kept as v3

pass	hypothesis	what was built	predeclared test	result	disposition
7	SAM2 can track a tail through a dark prefix	long one-sided propagation oracle with two negative controls	$\geq 3/4$ dark frames tracked, no drift in the controls	4/4; controls fail safe (mask empties)	evidence for pass 8
8	track until evidence of failure	v4: whole-run tracking, one-sided from the edges, stop on empty / no centreline / area $\geq 2.5\times$	$\Delta TP \geq 3$, ΔFP hidden = 0, no wrong-part on review	+5 TP, +8 FP: SAM2 slides onto a black leg and never empties	rejected
9	consensus can audit accepted frames	v3a: replace each accepted mask by the intersection of its neighbours' propagations	$\Delta FP \leq -1$, $\Delta TP \geq 0$, no new wrong-part	-2 FP, +2 TP — both were whole-cat boxes the truth had accepted (revision 1.2)	candidate, not default (it shaped the truth it gained on)
10	low-confidence partials are a truth problem	primary stratum = visible/partial with identity confidence high/medium; blinded re-review with ± 1 s of video	licking strip resolves or is excluded	unresolved; excluded (revision 1.3)	truth rule
11	v3 generalises	prospective holdout #1: four clips, metadata-only draw, truth before inference	$\Delta R > 0$, $\Delta P \geq -0.02$, no new repeated failure	ΔR 0, ΔP -0.012, a rump read as a tail for 17 frames	not met
12	—	note 05 v0.1	—	—	published
13	an existing SigLIP class already beats "tail" on the rump	offline margin audit of the six crop scores on every accepted frame	engineering screen: remove most of the Taiwan run, no new repeated failure, ≤ 2 recall points lost	parameter-free tail $> \max(\text{others})$: -31 dev TP, -40 holdout TP (a hanging cream tail is "leg")	rejected
14	the crop, not the model, is the problem	re-score on SAM2-mask-tight crops	Larry $\geq 30/40$ improve; Taiwan $\geq 12/17$ keep a negative margin; ≤ 5 dev flips	1/40; 14/17; 9 flips; AUROC 0.38 $\rightarrow$ 0.18	rejected; zero-shot verification dropped
15	v3 generalises (second draw)	holdout #2: next eight clips in the frozen order, truth before inference	as pass 11	ΔR +0.034, ΔP -0.008, no new repeated failure	met ; precision 0.73
16	mask geometry separates the false assertions	six features on 191 TP / 46 FP across all three sets	direction replicates on all sets, pooled AUROC ≥ 0.80 , ≥ 2 FP modes	max AUROC 0.67; direction reverses between sets	rejected; single-threshold geometry dropped
17	—	note 06 v0.1	—	—	published
18	the frozen embedding contains tail-vs-not-tail signal the text heads miss	linear probe on SigLIP embeddings of the 237 accepted crops, clip- and run-weighted, leave-two-clips-out over the 11 clips that have accepted frames (55 folds), threshold at 98 % of weighted training TPs	held-out FP -25 % with TP loss ≤ 2 %, on ≥ 3 clips, no clip of ≥ 10 TPs losing > 1	TP 191 $\rightarrow$ 44, FP 46 $\rightarrow$ 15; six clips lose more than one	rejected (exploratory: labels not human-checked)
19	the record has holes a reviewer would find	reviewer audit of scorer, truth rules, notes and manifests	—	scorer judged the detector's box, not the curve; visible-without-reference scored any assertion correct; prose counts disagreed with the manifests	scorer v2; one generated aggregate; notes 05 v0.3 and 06 v0.2

Table 2. Truth revisions.

version	pass	change	found by
1	4	first written-down truth: visibility, identity confidence, condition, reference box; reviewer is a model	—
1.1	8	second cat's tail recorded (a lt_*); scorer instance-agnostic; the tuxedo tail on the scratcher, f007–f017 of the jumping clip, annotated for the first time	the oracle in pass 7 tracking the other cat
1.2	9	walking f006/f007 references replaced (they were whole-cat detector boxes); containment rule made one-way	the audit in pass 9
1.3	10	primary stratum requires identity confidence high/medium; identity column; licking f027–f035 re-reviewed with temporal context and left unresolved	the reviewer model's rule
scorer v2	19	the reported curve is judged, not the detector's box; a collapsed centreline is unusable_curve; a visible tail without a reference box needs a hand judgement; holdouts scored as frozen and adjudicated	the reviewer's audit
—	15	<i>Andra and Billy</i> f012–f014 moved to uncertain after v1 drew a mask where the blind reading said hidden	method output; resolved down, never up

3. What the table says

Three things. First, the only change that survived is the one that adds no threshold and no new model: it asks two propagations to agree and refuses when they do not. Second, the truth moved four times, three of them because a method found something the annotator had not, and each time the fix was to make the truth stricter or to exclude, never to accept the method's reading. Third, the failures cluster: every semantic and structural verifier tried was a per-frame test of a crop or a mask, and every one failed because true tails and false assertions overlap on whatever single axis was chosen — a hanging cream tail is what the zero-shot model calls a leg, a curled grey tail is as body-attached as a whole-cat mask. The reviewer model's own reading at the end of pass 16 was to stop tuning here, and that is what this note does. Its audit at pass 19 then found that the scorer had been judging the detector's box rather than the curve the method reports, which hid six collapsed centrelines; the corrected numbers are in note 05 v0.3 and do not change any disposition in Table 1. One limitation stands over every number: the 603 frames come from 16 clips and are not 603 independent observations — one clip supplies 40 near-identical easy tails, another a 17-frame repeated mistake — so the aggregate also reports per-clip counts, a macro average over clips and the number of distinct false runs (seven on holdout #2, not 27).

4. What comes next

The reviewer model's proposal for the next phase was a linear probe on the frozen SigLIP image embedding of the existing crop, trained on the 237 accepted predictions (191 true, 46 false) with each clip and each contiguous same-label run given equal weight, evaluated by exhaustive leave-two-clips-out, threshold chosen on training clips only as the highest that keeps 98 % of weighted training true positives, and promoted only if held-out false assertions fall by 25 % with true ones down no more than 2 %, on at least three different held-out clips, with no clip of ten or more true frames losing more than one. It was proposed for 16 clips and 120 folds; only 11 clips have an accepted frame, so the executed experiment was 55 folds. It was run as an exploratory probe, because the labels are model-annotated and a human has not checked them (that item is filed and open), and it failed every clause: held-out true frames fell from 191 to 44 while false ones fell from 46 to 15. With one cat per clip, a probe trained on ten cats' tails does not recognise the eleventh's, which is note 04's result again at a larger scale. The frozen embedding is not sufficient at this data size; the sequence stops there rather

than escalating to fine-tuning.

Acknowledgements

As for note 05: the passes were a back-and-forth between Claude Code and ChatGPT; the truth was annotated by the former; the human pass is pending.